```
!pip install nltk pandas scikit-learn -q
```

```
import pandas as pd
import numpy as np
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity


nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')


# Load only the first 10,000 reviews
df = pd.read_csv(
    '/content/Reviews.csv',
    nrows=10000,
    on_bad_lines='skip'
)

print("Dataset Shape:", df.shape)
print(df.head(5))

# Select Text column
df = df[['Text']]

# Remove missing values
df = df.dropna()

# Remove duplicate reviews
df = df.drop_duplicates()

# Keep first 10,000 reviews
df = df.head(10000)

print("Number of reviews:", len(df))
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
Dataset Shape: (8187, 10)
   Id   ProductId          UserId                      ProfileName  \
0   1  B001E4KFG0  A3SGXH7AUHU8GW                       delmartian
1   2  B00813GRG4  A1D87F6ZCVE5NK                           dll pa
2   3  B000LQOCH0   ABXLMWJIXXAIN  Natalia Corres "Natalia Corres"
3   4  B000UA0QIQ  A395BORC6FGVXV                             Karl
4   5  B006K2ZZ7K  A1UQRSCLF8GW1T    Michael D. Bigham "M. Wassir"
```

```
   HelpfulnessNumerator  HelpfulnessDenominator  Score        Time  \
0                     1                       1      5  1303862400
1                     0                       0      1  1346976000
2                     1                       1      4  1219017600
3                     3                       3      2  1307923200
4                     0                       0      5  1350777600

                Summary                                               Text
0  Good Quality Dog Food  I have bought several of the Vitality canned d...
1      Not as Advertised  Product arrived labeled as Jumbo Salted Peanut...
2  "Delight" says it all  This is a confection that has been around a fe...
3         Cough Medicine  If you are looking for the secret ingredient i...
4            Great taffy  Great taffy at a great price.  There was a wid...
Number of reviews: 8158
```

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import numpy as np
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from sklearn.feature_extraction.text import TfidfVectorizer
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df = pd.read_csv(
    '/content/Reviews.csv',
    nrows=10000,
    on_bad_lines='skip'
)

df = df[['Text']]
df = df.dropna()
df = df.drop_duplicates()
df = df.head(10000)
print("Number of reviews:", len(df))


stop_words = set(stopwords.words('english'))
def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

df['Cleaned_Text'] = df['Text'].apply(preprocess_text)
for i in range(5):
    print("Review", i + 1)
    print("Original :", df.iloc[i]['Text'])
    print("Cleaned  :", df.iloc[i]['Cleaned_Text'])
    print("-" * 80)

vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(df['Cleaned_Text'])

print("\nTF-IDF Matrix Shape:", tfidf_matrix.shape)
```

```
print("Vocabulary Size:", len(vectorizer.vocabulary_))


def retrieve_reviews(query, top_n=5):


    cleaned_query = preprocess_text(query)


    query_vector = vectorizer.transform([cleaned_query])

    similarity_scores = cosine_similarity(
        query_vector,
        tfidf_matrix
    ).flatten()

    top_indices = similarity_scores.argsort()[-top_n:][::-1]

    print(query)
    print("=" * 100)

    for rank, index in enumerate(top_indices, start=1):

        print("\nRank:", rank)
        print(round(similarity_scores[index], 4))

        print(df.iloc[index]['Text'])

        print(df.iloc[index]['Cleaned_Text'])

        print("-" * 100)

queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]

for query in queries:
    retrieve_reviews(query)
```

```
Number of reviews: 9513
Review 1
Original : I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it s
Cleaned  : bought several vitality canned dog food products found good quality product looks like stew processed meat smells better labrador finicky appreciates product better
--------------------------------------------------------------------------------
Review 2
Original : Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the prod
Cleaned  : product arrived labeled jumbo salted peanutsthe peanuts actually small sized unsalted sure error vendor intended represent product jumbo
--------------------------------------------------------------------------------
Review 3
Original : This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then l
Cleaned  : confection around centuries light pillowy citrus gelatin nuts case filberts cut tiny squares liberally coated powdered sugar tiny mouthful heaven chewy flavorful highly recomm
--------------------------------------------------------------------------------
Review 4
```

```
Original : If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in addition to the Root Beer Extract I ordered (which was good) and made some
Cleaned  : looking secret ingredient robitussin believe found got addition root beer extract ordered good made cherry soda flavor medicinal
--------------------------------------------------------------------------------
Review 5
Original : Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.  If your a taffy lover, this is a deal.
Cleaned  : great taffy great price wide assortment yummy taffy delivery quick taffy lover deal
--------------------------------------------------------------------------------

TF-IDF Matrix Shape: (9513, 23530)
Vocabulary Size: 23530
great product with fast shipping
====================================================================================================

Rank: 1
0.5002
Enjoyed the product and they also provided very fast shipping.  I'm about out and need to order more.
enjoyed product also provided fast shipping im need order
----------------------------------------------------------------------------------------------------

Rank: 2
0.4431
The tea is good and fresh. We enjoy it. The shipping is fast and cost is reasonable.<br />I recommend this product.
tea good fresh enjoy shipping fast cost reasonablebr recommend product
----------------------------------------------------------------------------------------------------

Rank: 3
0.4007
The energy drink is a great product. The shipping price is crazy tho. I paid more for the shipping then both products together, so not a great deal.
energy drink great product shipping price crazy tho paid shipping products together great deal
----------------------------------------------------------------------------------------------------

Rank: 4
0.3973
My daughter lives in Hawaii and sent me some great coffee from there. I have a Keurig and  was not sure how to use the new coffee. I got these fast fast and now am using my Hawaiian coff
daughter lives hawaii sent great coffee keurig sure use new coffee got fast fast using hawaiian coffee great product
----------------------------------------------------------------------------------------------------

Rank: 5
0.3894
Fast shipping, items were packaged nicely and are as described. Also, very tasty :) Highly recommend this product to anyone looking to buy.
fast shipping items packaged nicely described also tasty highly recommend product anyone looking buy
----------------------------------------------------------------------------------------------------
disappointed
```